

Inspection and Quality Control Robots

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I. SIGNIFICANT METHODOLOGIES

Inspection and quality control robots spilt broadly into two families; vision-based systems that judge a part using image rendering, and contact-based systems that use physical touch to measure a part. The literature reviewed for this project leaned heavily toward vision system, since defect detection dominates in recent publications, but there were three papers that stood out for containing governing equations, physical hardware, and reported detailed results. Reference [18] solves inverse kinematics to drive a touch probe towards a contact point. Reference [15] shows how as few as five contact points can be enough to fully determine a cylinder's dimensions in the robot's base frame. Reference [17] gives us force control, keeping a probe in constant contact with a surface that is neither flat nor rigid. Together these three form the basis for the proposed methodology in Section IV, and Table 1 will compare them against the remaining literature. Each paper uses its own notations for variables for axis, angle, etc., thus, to keep the paper's originality, each notation will carry from the paper being used, instead of a shared symbol confusing each equation.

A. Automatic Contact-Based 3D Scanning Using an Articulated Robotic Arm [18].

Reference [18] uses a six degree of freedom arm that checks a part by touching it instead of taking a picture of it. A digitizer probe sits at the end effector, and it sends a signal the moment it touches something, and the controller turns that touch into a 3D coordinate. The problem here is basically inverse kinematics: if you know where you want the probe to be and how you want it angled, what six joint angles get you there.

The pose of the end effector is written as a 4x4 homogeneous transformation matrix 0T , where the rotation part and the position vector $p = (p_x, p_y, p_z)$ tell you where the probe tip is sitting in the base frame. Since the last three joints all meet at one point, the wrist center, that point gets solved for first, before touching the last three angles at all:

$$x_c = p_x - d_6 r_{13}, y_c = p_y - d_6 r_{23}, z_c = p_z - d_6 r_{33}$$

Once you have that, the first three joint angles come out of basic geometry, treating the arm like a two-link planar arm since you already know how far away the wrist center is:

$$\theta_1 = \alpha \tan 2 \left(\frac{y_c}{x_c} \right)$$

$$\theta_2 = \alpha + \beta$$

$$\theta_3 = \cos^{-1} \left(\frac{(l_2^2 + d_4^2 - R_1^2)}{2l_2 d_4} \right)$$

Alpha and beta come from the triangle formed by the link lengths and how far off the wrist center is vertically and horizontally. One small note: theta_1 is written here with

atan2 instead of the plain arctangent the paper uses, since atan2 actually gets the angle right in all four quadrants instead of just two, so it is a small fix on top of what [18] wrote, not a different answer. The last three angles orient the wrist and just fall out of the rotation matrix once the first three are plugged back in. This closed form matters for inspection because the arm has to hit a lot of contact points one after another, and it can't afford to numerically search for each one.

The arm itself is based on the open-source Annin Robotics AR3, a 1 kg payload, 600 mm reach arm run by stepper motors through planetary gears and controlled with an Arduino Mega 2560, with a Renishaw style kinematic coupling touch probe on the end. Once you tell it the size, position, and point density of the area you want scanned, it plans straight line paths between points on its own and spits out the finished scan as an STL file.

They checked the system in two ways. Following ASME B89.4.22, which is a standard for these kinds of measuring arms, a calibrated sphere test came out to an average diameter error of 0.036 mm, and a repeatability test on a single point found error going from plus or minus 0.0387 mm at 120 mm out to plus or minus 0.0712 mm at 500 mm, so accuracy gets worse the farther the arm reaches from its base, which makes sense. Separately, they scanned a 3D printed NACA 6409 airfoil ten times and compared the scan to the actual design file using something called Chamfer distance, basically a way to measure how far apart two clouds of points are on average and got 0.823 mm on average.

The system has one clear weak spot: it's open loop, meaning it has no way to sense or fix the tiny bit of arm deflection that happens right when the probe hits something, so a hard or sudden hit adds error that nothing catches downstream. It also can't reach overhangs or the underside of a part since it only comes at things from one side. This is meant for reverse engineering and checking part dimensions, and because it has a closed form inverse kinematics solution plus a real standardized accuracy test, it's the clearest example in the literature of Craig's kinematics chapters actually being used to build a contact-based inspection tool, which is why it's the anchor for the method proposed in Section IV.

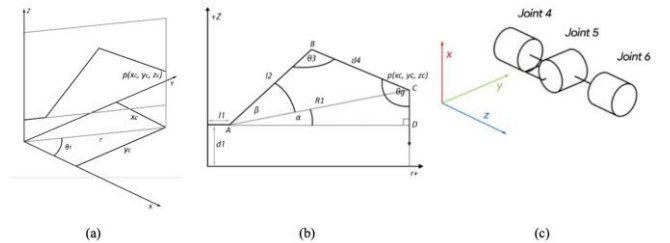


Figure 1. (a) Projection of arm on x, y plane. (b) Projection of arm on z, r plane. (c) Spherical joints [18]

B. Contact Point Determination for Pipe Inspection [15]

Reference [15] is ACES, a teleoperated, force-controlled six degrees of freedom arm built to look inside pipes. Instead

of scanning point by point like it does in [18], ACES asks a smaller question: how few points it takes to fully describe a cylinder, and how do you get those points without already knowing the pipe dimensions.

The main idea, straight from the paper, is that the arm touches five different spots on the cylinder, as long as no four of those five points sit on the same plane, and once you know where those five points are in the robot's base frame, that's enough to fully pin down the inner cylinder. The paper never actually writes this out as a formula, it's just stated as a geometry condition, so it's worth writing out here, and this part is our own formalization, not something taken from [15]: a cylinder needs five numbers to describe it, two for which way the axis points ($\hat{a}$), two for a point q sitting on that axis, and one for the radius r . Every contact point p_i gives you one constraint on those five numbers through how far it sits from the axis line:

$$|(p_i - q) - [(p_i - q) \cdot \hat{a}]\hat{a}| = r, i = 1, \dots, 5$$

if no four of the five points p_i are coplanar. Five of these give you just enough equations to solve for $\hat{a}$, q , and r , no more, no less. Once the cylinder is known, the arm traces a circle centered on that axis, hitting points about 50 mm apart, roughly 150 points per cross section, for the corrosion survey that comes after.

The arm rides on a Clearpath Jackal mobile base. The end effector has a contact sensor that finds each touch relative to the base frame, plus one of two swappable NDT probes, either a potential-field rod electrode or a single pulsed eddy current probe, whichever is used to actually check for corrosion once the geometry is known.

Since this paper is really a design and architecture study and not a hands-on trial, it doesn't report any dimensional accuracy of numbers. What it gives instead is a comparison of different in-pipe robot types, wheels, crawlers, helical, snake, inchworm, walking, and hybrid wall-pressing designs, judged on mobility, cost, and how hard they are to control.

The requirement that no four points be coplanar basically works as a built-in error check: break that rule and the cylinder can't be solved, so any real use of this method needs to check that the calibration points aren't degenerate before trusting the answer. The paper also points out that this kind of arm gets expensive and hard to control at elbows, tees, and places where the pipe diameter changes, sometimes needing a 7+ DOF arm to work around those spots. This is meant for corrosion checks on buried pipes like Steel Cylinder Concrete Pipe, somewhere you'd otherwise need teleoperation or to just dig it up. The five-point idea is the direct seed for the fitting method.

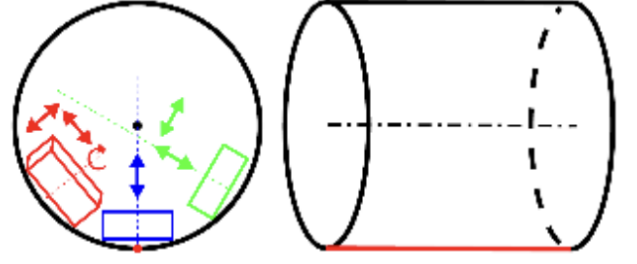


Figure 2. Robot position inside a pipe: cylinder front (left) and side (right) views [15]

C. Force-Controlled Contact Scanning for Co-Robotic Ultrasound [17]

Reference [17] deals with something [15] and [18] mostly skip what happens when the surface isn't rigid. Here the use case is mammography, where an ultrasound probe has to stay constant, even in contact with skin, and skin deforms and is never perfectly flat. Thus, the main problem is force control, not pure geometry, and the fix applies to any inspection task where the probe must track a surface that can flex while it's touching it.

The motion breaks into three steps. First, the robot moves the probe to a set distance above the target. Second, it moves straight down along the surface normal until a six-axis force and torque sensor senses the contact force hit 2 N, at which point it stops, which is basically a built-in safety check against hitting the surface too hard. Third, it does a linear scan, and this is where the actual governing equation comes in. The velocity of the probe splits into a part along the surface and a part into or away from it: $\vec{v} = v_t \tau + v_n \times \vec{n}$ (note: the overbars are hat)

Where tau-hat is the direction along the surface, n-hat is the direction straight out of the surface, v_t stays constant so the scan speed doesn't change, and v_n comes from a PID controller reacting to how far off the force is from the target:

$$v_n = K_p \left[e(t) + \frac{1}{T_i} \int e(\tau) d\tau + T_d \frac{de(t)}{dt} \right], \quad e(t) = F^* - F$$

F is the force actually being measured and F^* is the force you're aiming for. Basically, this law makes the probe push harder if the surface pulls away and back off if the surface pushes back, so the contact force stays close to constant while the tangential motion moves it across the part.

Hardware wise, it's a Universal Robots UR5 arm with a Robotiq FT-150 force and torque sensor between the flange and the probe, an Ultrasonix ultrasound unit with a linear array transducer, and a camera paired with ArUco markers to find the target. The software runs on ROS with an RRT planner to plan the path. Over twenty trials, the system held the contact force to a mean of 5.18 N with a standard deviation of 0.23 N against a 5 N target, overshooting by at most 0.72 N, and positional error stayed at 3.60 mm sideways and 7.70 mm along the depth direction, with every single trial finishing in one to two minutes.

The main weak spot is about 7 mm of positional repeatability, which is way too rough for actual dimensional checking, plus a single camera that can't give depth on its own

and an old test arm that the authors themselves suspected had some of its own joint error. As far as error handling goes, the PID loop is what keeps the contact force safe during the scan, but nothing corrects the arm's own drift, which is a totally separate problem from what the force controller is solving. This particular paper is about breast cancer screening, but the same force law works for any probe, ultrasound or not, moving across an industrial part that isn't perfectly rigid or flat.

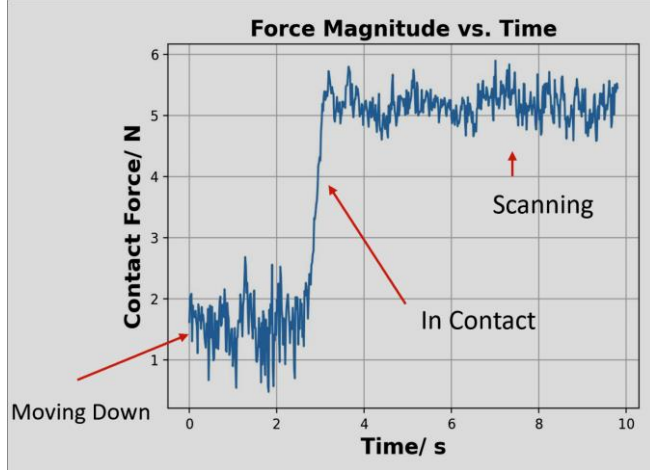


Figure 3. Force-vs-time graph showing the signal settling around the 5 N setpoint. [17]

D. Comparative Summary

Table 1. Comparison of featured contact-based inspection approaches. [15][16][17][18]

Ref	Method	Framework	Result	Limitations
[15]	Contact NDT probe, force-controlled 6-DOF arm	Five-point cylinder determination, no four coplanar	Design/architecture study; comparative platform evaluation	Underdetermined if points are coplanar; costly at elbows
[16]	Contact ultrasound probe, hybrid passive-active constant force	Positive/negative stiffness superposition	Force variation under 7.2%; range 12.3 to 21.3 N	Single mechanism cannot cover full clinical force range
[17]	Contact ultrasound probe, force-controlled 6-	Three-phase motion with PID normal-force control	Force held to 5.18 ± 0.23 N; ~ 7 mm repeatability	Repeatability too coarse for dimensional work; no depth sensing

	DOF arm			
[18]	Contact ultrasound probe, hybrid passive-active constant force	Positive/negative stiffness superposition	Force variation under 7.2%; range 12.3 to 21.3 N	Single mechanism cannot cover full clinical force range

Note: Added Reference [16] to increase comparative data.

II. PROPOSED METHODOLOGY

A. Vision-Based Surface Defect Inspection

This project introduces an automated robotic inspection and quality control system capable of detecting surface defects and inaccurate dimensions on manufactured components using an industrial robot with machine vision. Unlike a traditional fixed inspection station, the proposed system utilizes a six degree of freedom robotic arm armed with a high definition camera, laser displacement sensor and depth camera. The robotic arm allows the parts to be inspected from multiple angles, reducing mistakes, improving dimensional accuracy and allows the system to inspect components with complex geometries.

Traditional quality control methods rely on manual visual inspection performed by operators or a fixed inspection station. These approaches have been utilized widely throughout manufacturing, although they can suffer from inconsistent results due to operator error, limited viewing angles, and lighting conditions. Fixed camera systems also have a restriction on the complexity of geometries due to areas hidden from a single viewpoint. In order to overcome these limitations the proposed robotic inspection system uses a robotic manipulator that repositions sensors around the manufactured component to obtain multiple views. This will significantly improve defect detection allowing the system to examine difficult to reach surfaces.

The inspection process will begin when a manufactured part arrives at the inspection station on a conveyor belt. A proximity sensor will detect the arrival of the component and signal the robot controller to begin the inspection cycle. The robot will move to an initial viewing position where the camera will capture a high resolution image of the component. This image will then be ran through a convolutional neural network to analyze and identify visible defects such as dents, scratches, cracks, missing features, and surface marring.

If none of these imperfections are detected during the initial inspection, the robot will capture dimensional measurements using the laser displacement sensor and depth camera. If defects are identified, the robot automatically will reposition itself to additional viewpoints to verify any defects and eliminate false detections caused by shadows or reflections.

The motion of the robotic manipulator is determined using the Danvit-Hartenberg convention, which establishes the

geometric relationship between robot links. Each joint transformation is represented by a homogeneous transformation matrix, allowing the position and orientation of the end effector to be calculated throughout this inspection process. The overall transformation from the robot base to the end effector is expressed as

$$T_0^n = A_1 A_2 A_3 \dots A_n$$

Each transformation matrix is defined as

$$A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

These transformation matrices enable accurate positioning of the sensors relative to the component ensuring a repeatable measurement.

Forward kinematics are used to determine the position and orientation of the camera if the robot joint angles are known, while inverse kinematics calculates the required joint angles to move the camera to a desired inspection location. These relationships are shown as

$$T = f(q)$$

And

$$q = f^{-1}(T)$$

Q represents the vector of joint variables and T represents the desired end effector pose.

Following data collection, measurements from the high definition camera, depth camera, laser displacement sensors are combined through sensor fusion. Image processing algorithms extract surface features while dimensional measurements are compared against the CAD model of the component. Any deviation that exceeds manufacturing tolerances are automatically classified as a defect. The software then generates a report that can identify the defect, location, dimensions, and provide corresponding images. This information is stored in a quality control database to support future process improvement.

Verification of the proposed methodology will be performed by comparing the robotic inspection results with measurements obtained using a coordinate measuring machine and manual inspection. Multiple test parts that have known defects will be inspected to evaluate performance of the system. Statistical comparisons between robotic and traditional inspection methods will determine whether the proposed system provides equivalent or improved performance.

The proposed methodology has several advantages over traditional inspection methods. Automated robotic inspection can provide a more consistent measurement accuracy, increase production output, and eliminate operator fatigue. It can enhance worker safety by reducing a direct interaction with hazardous manufacturing equipment. The design also allows additional sensing equipment such as ultrasonic probes and thermal cameras to be integrated without significant modifications. Additionally, deep learning can allow the inspection system to improve over time as more data becomes available.

Despite these advantages, there are still limitations. The initial cost required for industrial robots and vision systems can be large for small manufacturers. Deep learning requires huge quantities of accurately described training data to achieve high accuracy and performance decreases under poor lighting conditions.

III. IMPLEMENTATION

The proposed robotic inspection and quality control system will be implemented using a combination of industrial robots, machine vision, and simulation software to create a fully automated inspection site. The implementation process will begin with the design of the robotic workcell in a modeling software before transitioning to a physical product. This will minimize implementation errors, reduce build time, and allow robot movement to be optimized before deployed.

The robotic platform selected for this project is a six degree of freedom industrial robot such as the FANUC LR Mate 200iD or ABB IRB 120. These robots provide positional accuracy, repeatability, and flexibility for any precise inspection task. Additionally, they are capable of reaching multiple viewpoints around the component. A custom mounting bracket will be attached to the robot end effector to hold the inspection sensors without affecting the robot performance. The primary sensing equipment consists of a high resolution camera for visual inspection, a depth camera for dimensional measurements and a laser displacement sensor for verifying tolerances. All together these sensors provide information that will allow both surface defects and dimensions to be detected during a single inspection cycle.

The inspection workstation will also include a conveyor system that transports manufactured components into the inspection area. A proximity sensor positioned at the end of the conveyor detects the arrival of the components and sends a signal to the robot controller to begin the inspection. After the inspection has been completed a programmable logic controller will direct the accepted parts toward the production line while defective components are diverted into a rejected bin for additional inspection or rework. This automated sorting process helps to reduce delays and minimizes operator involvement.

Several softwares will be used to develop, simulate, and operate the inspection system. Solidworks will be used to create CAD models of the inspected components and the robotic workcell. These CAD models will be imported into robot simulation software, such as FANUC RoboGuide or ABB RobotStudio where collision detection, reachability, and robot path optimization will be performed before it is physically implemented. MATLAB combined with the robotic toolbox will be used to verify forward and inverse kinematic calculations and analyze robot trajectories. Python will be utilized as the primary language for machine vision, because it has a large amount of support for image processing and artificial intelligence. OpenCV will perform image processing while a convolutional neural network will classify defects automatically. Communication between the robot controller, vision system, and PLC will be connected using Ethernet to ensure fast real time data exchange.

Flow chart:

1. Component arrives at inspection station
2. Proximity sensor detects component
3. Robot moves to home position
4. Robot moves to initial inspection position
5. Camera captures an image
6. Deep learning model analyzes the image for defects
7. Robot moves to additional viewpoints if defects are suspected
8. Laser displacement sensor and depth camera perform dimensional measurements
9. Sensor fusion combines all data
10. Measurements are compared with CAD tolerances
11. PLC sorts the component into accepted or rejected paths
12. Inspection results are stored within the database

The implementation will first be checked through a simulation before any physical testing begins. RoboGuide can ensure that all robot trajectories don't pose a risk of collision and every inspection position is reachable. MATLAB simulations will verify the correctness of forward and inverse kinematic calculations and confirm that the manipulator follows smooth trajectories. Once the virtual model has been verified the system will be tested on a physical robot for testing.

Although this proposed methodology is intended for a large scale project it has been designed with scalability in mind. The architecture allows adding additional sensing technologies, including thermal cameras and ultrasonic probes to be integrated without major modifications. This inspection system is easily repeatable to be included on any machine in a manufacturing shop. Future improvements may include additional deep learning once more data is collected. Technology will enable the robot to optimize its trajectory and learn different angles to better inspect complex parts. These advancements will improve its detection accuracy and reduce inspection times, making the proposed system suitable for modern manufacturing environments.

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